



CMPG325 Ngaka Sports

CMPG325 Network Design Project (CLI-119)

Client Design Review

Ngaka Sports Development Foundation — Mahikeng

The CMPG325 Network Design Project by CLI-119 will give a comprehensive Packet Tracer design for the Ngaka Sports Development Foundation in Mahikeng. This includes an IP addressing scheme of 10.46.0.0/16, VLAN configuration, and full default routing to the ISP. It takes care of budget and performance by utilizing existing switches and solving the problem of CR8 with regard to shared printing.

Project Information

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Project ID	CMPG325-2026-119
Client ID	CLI-119
Addressing Block	10.46.0.0/16
Assigned Challenge	Default Routing (Edge/ISP Path Design)

1. Client Requirements

1.1 Organisation Structure & Departments

Assumption: The organization has offices in one location in Mahikeng with 5 major departments:

- 1.Admin and Management
- 2.Finance and HR
- 3.Sports Programs and Coaching
- 4.Facilities and Events
5. IT Support

Justification: This organizational design structure makes perfect sense for a sports development body in a region. There are no details on any departments provided in the case scenario.

1.2 Users, Devices & Subnetting

Assumption: 15 to 30 people per department, one VLAN per subnet. One printer is shared by the Finance and Sports Program departments (the departments are unable to print).

Justification: The scale is appropriate to the type of organization. The use of subnets /24 in the allocated /16 block gives simplicity and scalability.

1.3 Equipment & Budget Constraint

Assumption: 1 edge router, 1 core router, 2 existing Layer 2 switches (reuse). No additional switches purchased.

Justification: Satisfies the constraint: "Limited equipment budget – use existing switches if possible."

1.4 Change Request CR8 — Shared Printer

Assumption: Finance and Sport Programs belong to two distinct subnets. By default, VLAN20 - Finance and VLAN30 - Sports Programs cannot share a printer. The Printer belongs to the VLAN 50 Shared Printers. This will be done through routing.

Justification: Meets the requirement directly: "A printer pool area has to support two departments which presently do not have printing facilities."

1.5 Default Routing Design

Assumption: Default route in Edge Router points to the ISP. Default route in Core Router points to the Edge Router.

Justification: Designing of the standard edge/ISP pathway fulfills the networking assignment requirements.

2. Physical Topology

The network utilizes a simple three-layer network topology, which consists of an ISP connection, Edge Router that provides outside connectivity, a Core Router that handles inside routing, and two previously available Layer 2 switches that have been used because of budget issues. Switch 1 will be used for Admin, IT, and Guest networks, while Switch 2 will be responsible for Finance, Sports Programs, Facilities, and printer areas.

The physical design consists of:

1 Edge Router — connects to simulated ISP

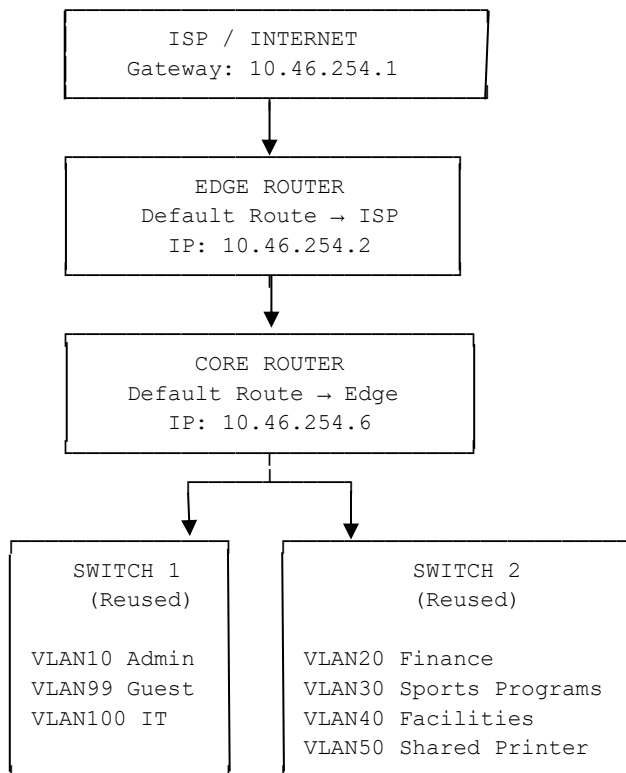
1 Core Router — internal central router

2 Reused Layer 2 Switches — no new switches purchased

Cabling: Copper straight-through connections between all devices

Switch 1 to Admin, IT, Guest devices

Switch 2 to Finance, Sports Programs, Facilities, Shared Printer



3. Logical Topology

Departmental traffic is segregated using VLANs, with each VLAN getting a dedicated /24 subnet.

VLAN Assignments:

- VLAN 10 — Admin and Management
- VLAN 20 — Finance and HR
- VLAN 30 — Sports Programs
- VLAN 40 — Facilities and Events
- VLAN 50 — Shared Printer Zone
- VLAN 99 — Guest Network
- VLAN 100 — IT Support

Routing Strategy:

- The default route used by Core Router directs traffic to Edge Router.
- Edge Router utilizes the default route that goes to the ISP.
- Core Router is responsible for providing connectivity between different VLANs, allowing both VLAN 20 – Finance and VLAN 30 – Sports Programs to use Shared Printer from VLAN 50.
- All internal traffic goes to the Internet via the Edge Router.

4. IP Addressing Plan

VLAN	Department / Purpose	Subnet	Gateway
10	Admin & Management	10.46.10.0/24	10.46.10.1
20	Finance & HR	10.46.20.0/24	10.46.20.1
30	Sports Programs	10.46.30.0/24	10.46.30.1
40	Facilities & Events	10.46.40.0/24	10.46.40.1
50	Shared Printer Zone	10.46.50.0/24	10.46.50.1
99	Guest Network	10.46.99.0/24	10.46.99.1
100	IT Support	10.46.100.0/24	10.46.100.1
—	Edge to ISP Link	10.46.254.0/30	ISP: 10.46.254.1 / Edge: 10.46.254.2
—	Edge to Core Link	10.46.254.4/30	Edge: 10.46.254.5 / Core: 10.46.254.6

Printer IP: 10.46.50.10/24

5. Initial GitHub repository

Repository created: CMPG325-CLI119-NgakaSports

Repository Link: <https://github.com/RikhotsoM/CMPG325-CLI119-NgakaSports>

README.md uploaded containing: project info, assumptions, topology description, IP addressing plan

MIT License added

The designed solution will segregate traffic belonging to different departments using VLANs and /24 subnets, keeping the current Layer 2 switches because of the budgetary considerations. The Core Router will be the internal router path, and the Edge Router will be the external ISP routing path by means of the default route. VLAN 50 can be used for reaching the IP address 10.46.50.10 through CR8.